PV102 Day 5: Battery Storage Autonomy and Critical Loads

Students decide what the battery is for, then estimate storage need from critical loads, runtime, autonomy days, voltage, usable capacity, and safety boundaries.

PV102 | CTS Day 21 | Instructor Run Sheet
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Core rule

Storage has a defined job: carry named loads through named time.

Learning targets

- Explain the difference between battery capacity, usable capacity, and autonomy.
- Estimate storage need from essential daily Wh and autonomy goal.
- Identify why battery chemistry, BMS limits, temperature, ventilation, and overcurrent protection matter.
- Defend load shedding as a design tool, not a failure.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Battery spec cards Load tables from Day 3 Calculators Scenario cards Storage worksheet Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 4 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work triggers.

1:00-2:15	Critical Load Storage Plan	Teams reduce an oversized wish list into a critical-load storage plan that a real system could support.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 6 connects storage and array decisions to inverter, charge controller, distribution, and protection boundaries.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use battery storage autonomy and critical loads to make the next design decision safer and more defensible. The rule is simple: Storage has a defined job: carry named loads through named time."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Start with the Day 3 load table or provided scenario.
- Mark each load as critical, useful, or optional.
- Calculate critical daily Wh.
- Choose an autonomy target and estimate required Wh before adjustment.
- List the battery information required before final sizing.
- Handoff the critical-load list, autonomy target, and rejected optional loads.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 6 connects storage and array decisions to inverter, charge controller, distribution, and protection boundaries.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.